FMRI-CPCA

Last Updated: November 20, 2023

This manual is intended to complement the fMRI-CPCA tutorial manual. The tutorial manual provides a good basic overview of how to run fMRI-CPCA. This manual includes supplementary information to review before conducting an actual analysis.

File Organization

When running CPCA, it is preferable to do so on a local Linux computer rather than the SAN drive. This is because the SAN drive has limited storage and will drastically increase computation time.

On the local computer, it is important to create an analysis folder for your results within your study folder with the data. When launching CPCA from MATLAB, ensure that the path is set to this analysis folder so that your progress will be saved in this location and not elsewhere on the computer.

There are several ways to organize the fMRI data folders. If you are conducting CPCA with only one group (i.e. controls or patients) and one condition per study, then your data should be organized into folders in the following manner: Study/Subjects/Runs. If you have various groups, your data should be organized as Study/Groups/Subjects/Runs. If each subject only has one run folder, then you do not necessarily need to specify a run folder.

Note: if separation of 4D NIFTI files into individual 3D volumes was to be performed, then ensure that the individual 3D volumes were numbered such that their orders and organizations were not changed. For instance, ensure the individual 3D volumes are numbered like 001, 002, 003, etc., as opposed to being numbered like 1, 2, 3, etc.

Should you ever need to reorganize your fMRI data, consider symbolic links rather than copying and pasting. While you can copy and paste your data from different locations into folders, it is much quicker to organize your data by creating links to remote folders where your data is originally located. This can be done with a few lines of code in the terminal. Note that any changes you make to the data accessed from the remote folders will also be made in the original folders and vice versa. The link has to be between folders on the computer drive and not the SAN drive.

Data Linking Instructions

1. Create the parent directories for the links.
    - Create a folder named Data.
    - Within the Data folder, create a folder for each group (i.e. controls, patients).
2. Open Terminal.
3. Create the link by modifying target and link in the following lines of code.

    If you are using Linux: **ln -s target link**

    If you are using Windows: **mklink /J link target**

    - *Target* indicates where your original data is located.

- Modify target to be the entire path from the base folder of your computer to your original data subject folders for the same group selected in step 2.
- You may use a '*' to select all the subject folders within the group folder. A '*' is a wildcard operator, indicating any number or variation of characters may be in its place. If any of your folders include a space in the name, this space will need to be replaced by a '*'.
- For example, /home/cnoslab-ra/People/Chantal/Raw*/Healthy/* would be the appropriate target for all subject folders under Healthy for a specific study. An individual link will be created for each subject folder. The folder after Chantal contained a space in the title ('Raw Data'), so it was selected by including a '*' in place of the space.

- *Link* indicates the folder where you would like the link to appear.
  - Modify link to be the entire path from the base folder of your computer to the group folder where you would like to insert the linked subject folders. Ensure that the parent directories are already set up for you to select.
  - If any of your folders include a space in the name, this space will need to be replaced by a '*'.
  - For example, /home/cnoslab-ra/People/Chantal/MS_TGT/Data/Healthy would be the appropriate link for the location you would like to put all healthy subject folders.

After modifying our linux code in our example, it should read:

ln -s /home/cnoslab-ra/People/Chantal/Raw*/Healthy/* /home/cnoslab-ra/People/Chantal/MS_TGT/Data/Healthy

This will create individual subject folder links in MS_TGT/Data/Healthy to the subject folders for a study's data, which is found under: Raw Data/Healthy/*.

After executing this code, the linked subject folders should appear in the group folder. They will be indicated as a linked folder by a black arrow.

4. Repeat step 3 for the other study subjects classified in the same group.
5. Repeat steps 3 and 4 for each group. Overall, you will need to execute a line of code for each study within each group.

Below is an example of the code needed to link subject folders from two different studies into group folders for the MS_TGT merged study.

```
cnoslab-ra@cnoslab-ra:~$ ln -s /home/cnoslab-ra/People/Chantal/TGT/Raw_Data/Func
tional/Healthy/* /home/cnoslab-ra/People/Chantal/MS_TGT/Data/Healthy
cnoslab-ra@cnoslab-ra:~$ ln -s /home/cnoslab-ra/People/Chantal/TGT/Raw_Data/Func
tional/Schizophrenia/* /home/cnoslab-ra/People/Chantal/MS_TGT/Data/Schizophrenia

cnoslab-ra@cnoslab-ra:~$ ln -s /home/cnoslab-ra/People/Chantal/Raw*/Healthy/* /h
ome/cnoslab-ra/People/Chantal/MS_TGT/Data/Healthy
cnoslab-ra@cnoslab-ra:~$ ln -s /home/cnoslab-ra/People/Chantal/Raw*/Schizophreni
a/* /home/cnoslab-ra/People/Chantal/MS_TGT/Data/Schizophrenia
```

Mask Checking

After you have created your mask, it is important to check that the mask includes all desired regions of the brain and did not crop any important regions. This process is slightly different for the internal and public versions of CPCA. The internal version is accessible through the CNoS Lab, while the public version is the version that may be downloaded online.

When the mask is created, each subject may remove some voxels from the ultimate mask. If there are mask issues, it may be due to a couple subjects removing too many voxels.

Internal Version of CPCA

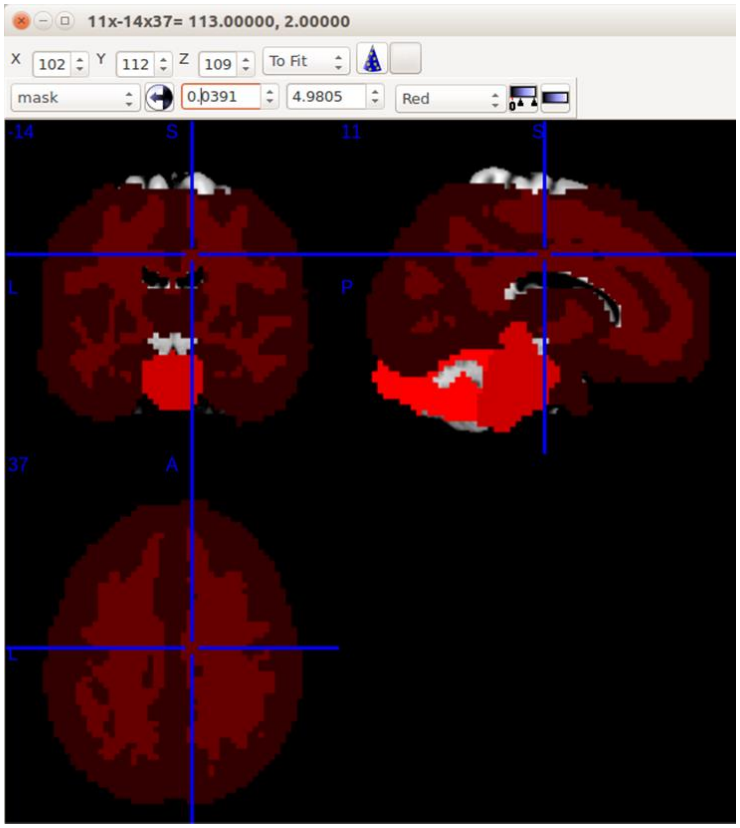
1. Visually inspect the mask with *MRIcron*.

   Open *MRIcron* and select the template under *File >Open templates >ch2bet.nii.gz*.

   Add your mask as an overlay on the template brain. Go to *Overlay > Add >* Select your mask file (mask.img).

   Your mask should now be overlaid on top of the template brain. Move the crosshairs to inspect all regions of the mask. Ideally, the mask will encompass nearly all of the template brain. Visually check if any regions are missing from the mask.

   In the below example, the mask was overlaid onto the ch2bet.nii.gz template. In this case, the top of the mask appears to be cut off. Careful inspection in the following steps may indicate why the mask is missing this region.



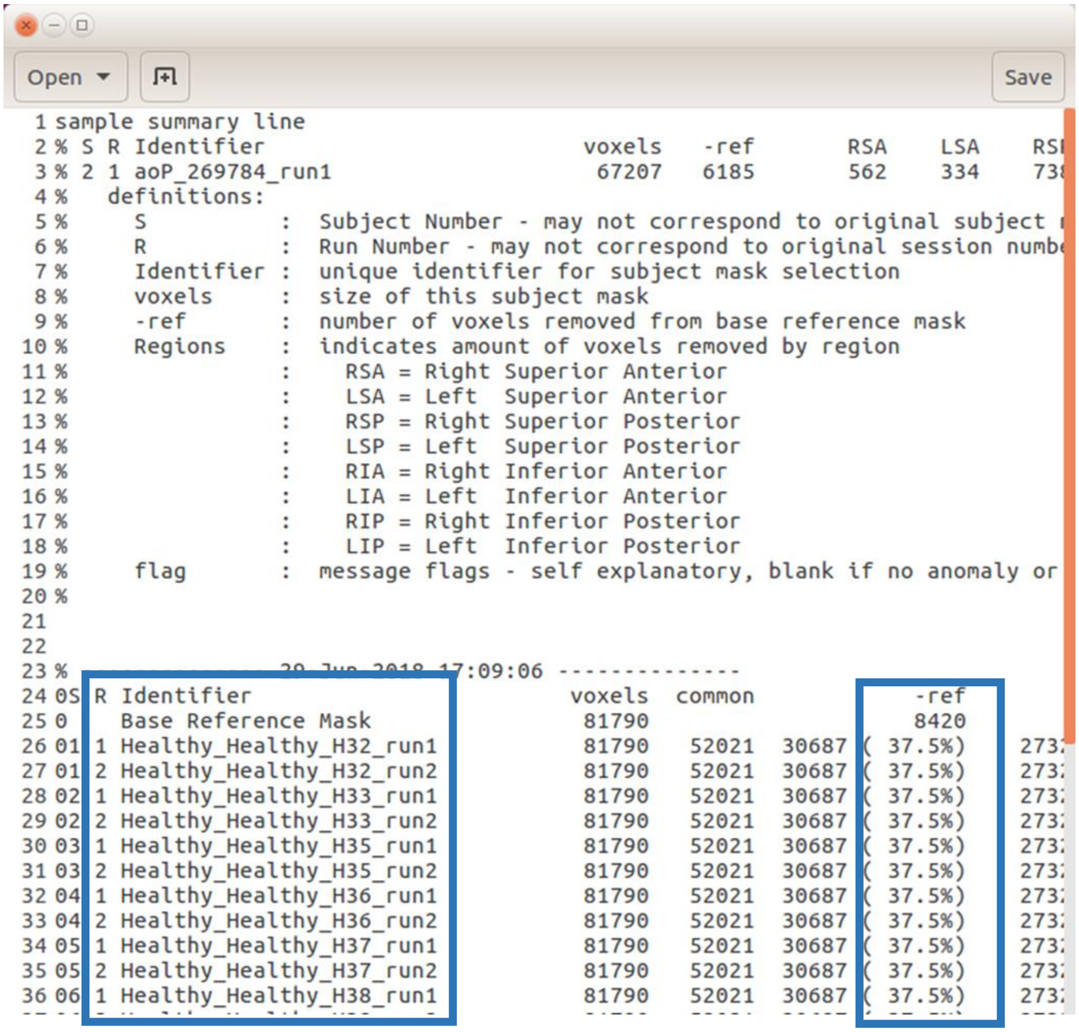
2. Check the mask verification files.

   Enter your analysis folder and select the *mask_logs* folder. Select *mask_creation_summary.log*. This text document contains information on the mask creation for each subject and each run.

   The first column indicates the subject number and run.

   The *-ref* column indicates the number of voxels removed from the base reference mask, as well as the percentage removed. This number should be fairly consistent across all subjects and all runs. The value of this number should likely fall between 30-40%. Scan through this column for any inconsistencies.

   Find the subject run with the largest percent removed. We will inspect the individual subject run mask of this participant, as they made the most significant contribution to removing voxels from the mask. If the mask was cut off in any way, checking this individual subject's mask is a good place to start to determine why part of the mask was cropped.

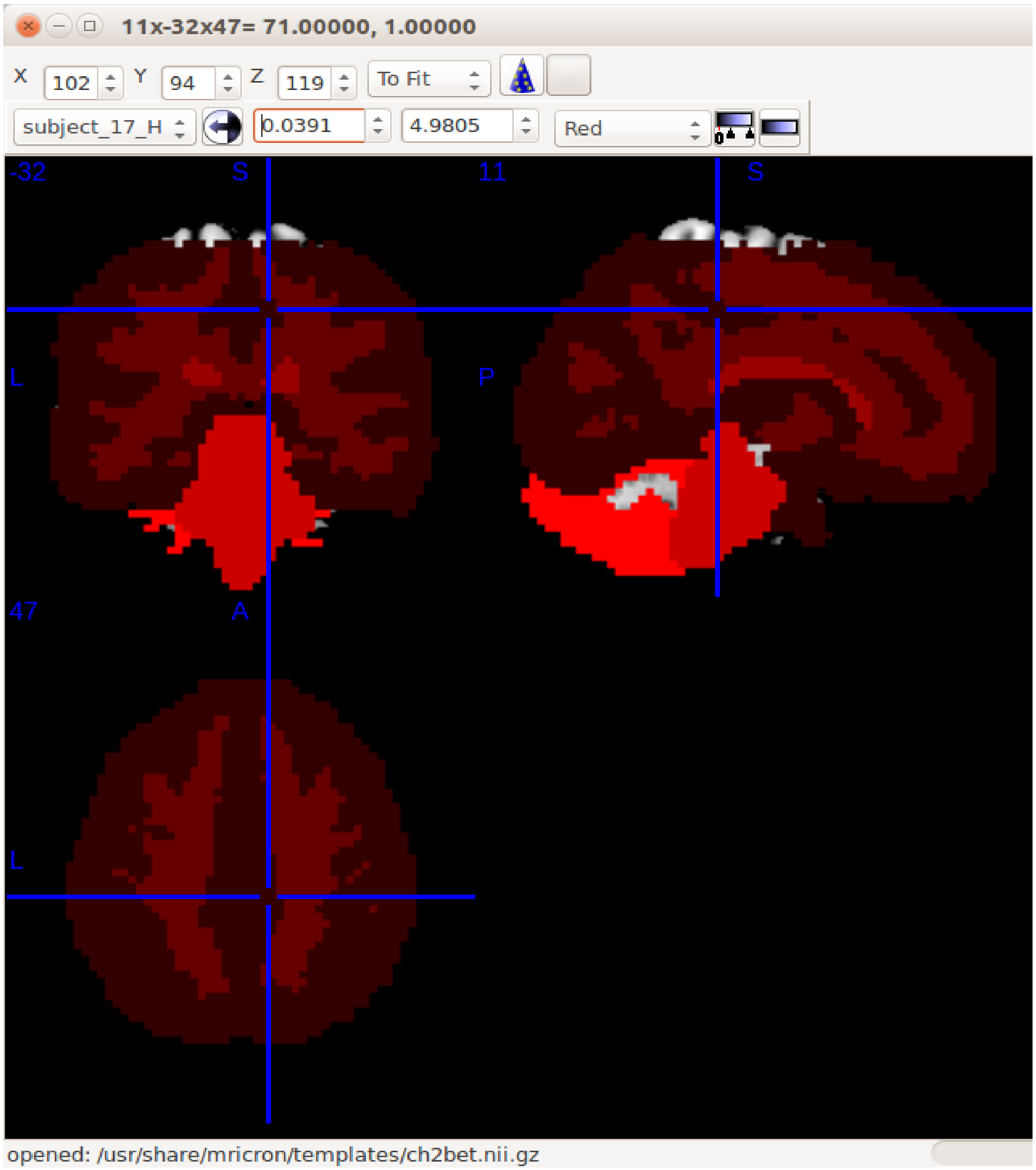


```
Open ▾                                                          Save
sample summary line
% S R Identifier                        voxels   -ref        RSA    LSA    RS
% 2 1 aoP_269784_run1                    67207   6185        562    334    73
%   definitions:
%     S            :  Subject Number - may not correspond to original subject
%     R            :  Run Number - may not correspond to original session numb
%     Identifier   :  unique identifier for subject mask selection
%     voxels       :  size of this subject mask
%     -ref         :  number of voxels removed from base reference mask
%     Regions      :  indicates amount of voxels removed by region
%                  :    RSA = Right Superior Anterior
%                  :    LSA = Left  Superior Anterior
%                  :    RSP = Right Superior Posterior
%                  :    LSP = Left  Superior Posterior
%                  :    RIA = Right Inferior Anterior
%                  :    LIA = Left  Inferior Anterior
%                  :    RIP = Right Inferior Posterior
%                  :    LIP = Left  Inferior Posterior
%     flag         :  message flags - self explanatory, blank if no anomaly or
%


%                 20 Jun 2018 17:09:06 --------------
0S R Identifier                        voxels  common           -ref
0    Base Reference Mask                81790                   8420
01 1 Healthy_Healthy_H32_run1           81790   52021   30687 ( 37.5%)    273
01 2 Healthy_Healthy_H32_run2           81790   52021   30687 ( 37.5%)    273
02 1 Healthy_Healthy_H33_run1           81790   52021   30687 ( 37.5%)    273
02 2 Healthy_Healthy_H33_run2           81790   52021   30687 ( 37.5%)    273
03 1 Healthy_Healthy_H35_run1           81790   52021   30687 ( 37.5%)    273
03 2 Healthy_Healthy_H35_run2           81790   52021   30687 ( 37.5%)    273
04 1 Healthy_Healthy_H36_run1           81790   52021   30687 ( 37.5%)    273
04 2 Healthy_Healthy_H36_run2           81790   52021   30687 ( 37.5%)    273
05 1 Healthy_Healthy_H37_run1           81790   52021   30687 ( 37.5%)    273
05 2 Healthy_Healthy_H37_run2           81790   52021   30687 ( 37.5%)    273
06 1 Healthy_Healthy_H38_run1           81790   52021   30687 ( 37.5%)    273
```

3. Inspect individual subject mask.

   Open *MRIcron* and select a template under *File > Open templates >ch2bet.nii.gz*.



Add your the individual subject mask (participant determined in step 3) as an overlay on the template brain. Go to *Overlay > Add > Select* your mask file. Individual subject masks are located under the *subject_masks* folder.

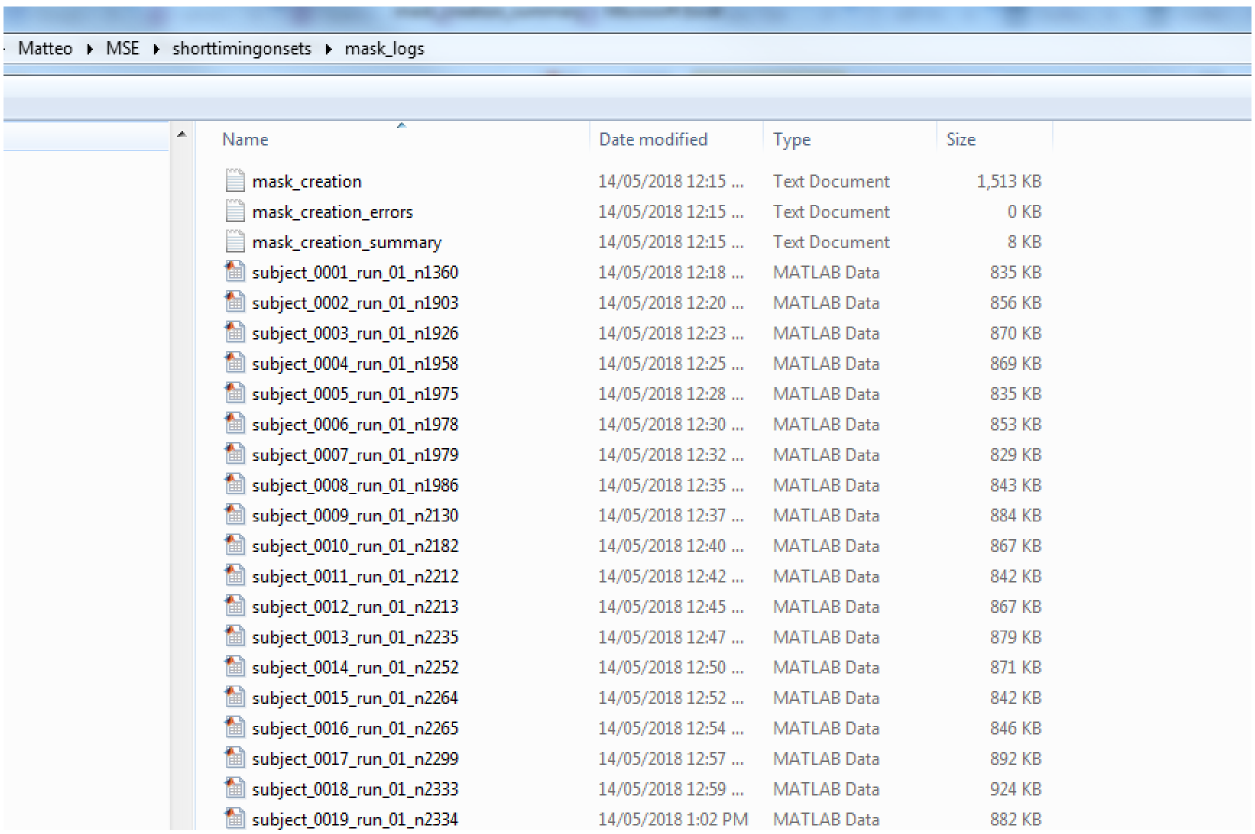
Your individual subject mask should now be overlaid on top of the template brain. Move the crosshairs to inspect all regions of the mask. Ideally, the mask will encompass nearly all of the template brain. Visually check if any regions are missing from the mask, and if these regions correspond to any missing regions from the overall mask.

4. Check subjects masks

   Check through any individual subject masks with relatively large percentages removed. If they appear to have removed an important portion of the mask, several things should be checked.

Public Version of CPCA

1. Enter the *mask _logs* folder.



2. Open the *mask_creation_summary.log* file in Excel. Examine the values beside “Adjusted”. These indicate how much each subject sequentially removed from the final mask. Flag the largest values, as these subjects had the largest impact on removing mask voxels. Note that the first subject should remove a large number of voxels.

8	%												
9	n1360	run	1	Mask:	198885	200475	removed:	394010	average:	393108	Adjusted:	396356	196539
10	n1903	run	1	Mask:	184206	187705	removed:	408689	average:	406355	Adjusted:	22010	174529
11	n1926	run	1	Mask:	186684	188502	removed:	406211	average:	405323	Adjusted:	5205	169324
12	n1958	run	1	Mask:	202038	203474	removed:	390857	average:	390257	Adjusted:	854	168470
13	n1975	run	1	Mask:	191419	192622	removed:	401476	average:	400709	Adjusted:	1415	167055
14	n1978	run	1	Mask:	207251	208743	removed:	385644	average:	384842	Adjusted:	336	166719
15	n1979	run	1	Mask:	179584	180804	removed:	413311	average:	412607	Adjusted:	2922	163797
16	n1986	run	1	Mask:	179983	180918	removed:	412912	average:	412430	Adjusted:	5046	158751
17	n2130	run	1	Mask:	210210	212857	removed:	382685	average:	380789	Adjusted:	33	158718
18	n2182	run	1	Mask:	192321	193777	removed:	400574	average:	399780	Adjusted:	329	158389
19	n2212	run	1	Mask:	186704	188804	removed:	406191	average:	405275	Adjusted:	907	157482
20	n2213	run	1	Mask:	203141	204542	removed:	389754	average:	388968	Adjusted:	322	157160
21	n2235	run	1	Mask:	211747	213374	removed:	381148	average:	380321	Adjusted:	1	157159
22	n2252	run	1	Mask:	197631	199729	removed:	395264	average:	394091	Adjusted:	111	157048
23	n2264	run	1	Mask:	191705	193382	removed:	401190	average:	400487	Adjusted:	92	156956
24	n2265	run	1	Mask:	212714	214372	removed:	380181	average:	379184	Adjusted:	82	156874
25	n2299	run	1	Mask:	198566	200960	removed:	394329	average:	393242	Adjusted:	184	156690
26	n2333	run	1	Mask:	212486	214188	removed:	380409	average:	379414	Adjusted:	3	156687
27	n2334	run	1	Mask:	183505	184730	removed:	409390	average:	408829	Adjusted:	574	156113
28	n2345	run	1	Mask:	212684	213840	removed:	380211	average:	379708	Adjusted:	37	156076
29	n2770	run	1	Mask:	190141	192028	removed:	402754	average:	401569	Adjusted:	240	155836
30	n2838	run	1	Mask:	186238	187121	removed:	406657	average:	406219	Adjusted:	620	155216
31	n3032	run	1	Mask:	191595	193618	removed:	401300	average:	400125	Adjusted:	197	155019
32	n3090	run	1	Mask:	178832	179936	removed:	414063	average:	413553	Adjusted:	811	154208
33	n3144	run	1	Mask:	195359	196456	removed:	397536	average:	397071	Adjusted:	23	154185

3. In the *mask_creation* file, look to see on which exact scan for the subject(s) the problem occurred.

n2213	run:	1	image:	1	Mask:	203918	removed:	388977	232
n1903	run:	1	image:	166	Mask:	187239	removed:	405656	233
n3470	run:	1	image:	1	Mask:	203902	removed:	388993	247
n3470	run:	1	image:	206	Mask:	201225	removed:	391670	258
n1978	run:	1	image:	1	Mask:	207385	removed:	385510	282
n1903	run:	1	image:	3	Mask:	187026	removed:	405869	288
n2006	run:	1	image:	1	Mask:	204038	removed:	388857	324
n1903	run:	1	image:	168	Mask:	185058	removed:	407837	329
n1903	run:	1	image:	202	Mask:	185055	removed:	407840	355
n3152	run:	1	image:	126	Mask:	197487	removed:	395408	355
n2783	run:	1	image:	1	Mask:	198904	removed:	393991	380
n2334	run:	1	image:	1	Mask:	184341	removed:	408554	421
n3234	run:	1	image:	1	Mask:	203320	removed:	389575	453
n1903	run:	1	image:	203	Mask:	184206	removed:	408689	504
n2838	run:	1	image:	1	Mask:	186558	removed:	406337	522
n3090	run:	1	image:	1	Mask:	179232	removed:	413663	591
n1958	run:	1	image:	1	Mask:	202769	removed:	390126	620
n3306	run:	1	image:	1	Mask:	201980	removed:	390915	694
n2212	run:	1	image:	1	Mask:	187015	removed:	405880	739
n1831	run:	1	image:	1	Mask:	191525	removed:	401370	820
n3305	run:	1	image:	1	Mask:	202520	removed:	390375	826
n3165	run:	1	image:	1	Mask:	187048	removed:	405847	968
n1903	run:	1	image:	167	Mask:	185603	removed:	407292	1103
n1975	run:	1	image:	1	Mask:	191687	removed:	401208	1211
n2087	run:	1	image:	1	Mask:	172968	removed:	419927	1986
n1979	run:	1	image:	1	Mask:	180210	removed:	412685	2453
n1926	run:	1	image:	1	Mask:	187305	removed:	405590	4440
n1986	run:	1	image:	1	Mask:	180232	removed:	412663	4625
n1903	run:	1	image:	1	Mask:	186402	removed:	406493	17463
n1360	run:	1	image:	1	Mask:	199611	removed:	393284	393284

In this example, it looks like n1903 took off quite a bit on image 166, 3, 168, 202, 203. Open these scans in *MRIcron* to see if there appears to be any problems.

Re-Creating the Mask

Once you have determined which subject(s) and scan(s) are liekly removing important voxels from the mask:

1. Verify that all of the preprocessing steps were successful for this subject. Redo preprocessing if necessary and then create the mask again.

2. If preprocessing was successful for this participant but their mask is still cut off, you may want to consider removing them from the dataset to improve your mask.

G & Timing Onsets

Before creating the G matrix in CPCA, it is a good idea to check over the timing onsets file that will be imported, especially if it has not previously been used before in another analysis.

When creating G, always check the number of bins to include in an analysis with Todd. The number of bins depends on the TR of your dataset.

Timing Onsets

A few items to consider:

1. If there are errors in the responses, you should probably only include correct responses, but check with Todd first.

2. If a subject has no responses for a timing vector, they will need to be excluded. Even though CPCA will run, this subject will not have values for the HDR shape.

3. It is important to note that two timing vectors cannot be subsets of one another.

4. If a certain subject does not have timing onsets for one of the conditions, do not include this as a blank timing onset in the timing onsets file. Instead, when creating G, manually deselect this condition for this participant such that the timing onsets template does not include it.

5. Check the number of onsets for each subject run condition

   The number of onsets for each subject run condition can easily be checked in MATLAB. If the data was entered correctly this number should be consistent with the experimental design across all subjects.

   Copy and paste the timing onset file into MATLAB. Type “whos” and press enter. This will produce a list of all the rows in your timing onset file, and the Size column indicates how many values are in each timing onset row. Ensure the number after 1x_ is consistent and as expected for each row.

Command Window

```
% --- timing onsets for subject 78 (n3165)
% -------------------------------------------------------
n3165_functional_Condition_1 = [ 14.9999333929 16.9999333929 18.9999333929 20.9
n3165_functional_Condition_2 = [ 55.002066674 57.002066674 59.002066674 61.0020


% -------------------------------------------------------
% --- timing onsets for subject 79 (n3231)
% -------------------------------------------------------
n3231_functional_Condition_1 = [ 15.0027666786 17.0027666786 19.0027666786 21.0
n3231_functional_Condition_2 = [ 54.997990562 56.997990562 58.997990562 60.9979


% -------------------------------------------------------
% --- timing onsets for subject 80 (n3455)
% -------------------------------------------------------
n3455_functional_Condition_1 = [ 14.9983667148 16.9983667148 18.9983667148 20.9
n3455_functional_Condition_2 = [ 55.0004762293 57.0004762293 59.0004762293 61.0



>> whos
  Name                                  Size              Bytes  Class     Attributes

  n1312_functional_Condition_1          1x48                384  double
  n1312_functional_Condition_2          1x48                384  double
  n1325_functional_Condition_1          1x48                384  double
  n1325_functional_Condition_2          1x48                384  double
  n1360_functional_Condition_1          1x48                384  double
  n1360_functional_Condition_2          1x48                384  double
  n1427_functional_Condition_1          1x48                384  double
  n1427_functional_Condition_2          1x48                384  double
  n1454_functional_Condition_1          1x48                384  double
```

typing their Name (ex. n1312_functional_Condition_1) and hitting enter.

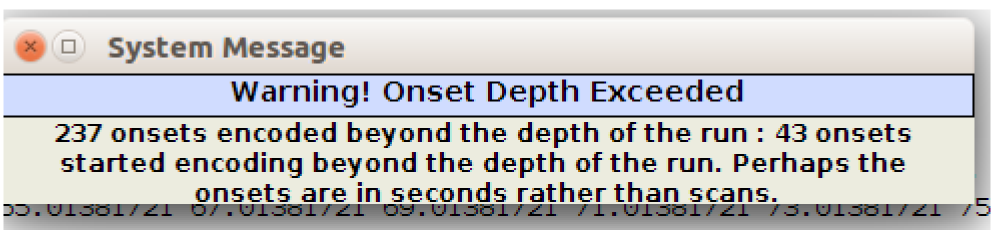
6. Quickly inspect the onset file to ensure the values make sense.

    Verify that the number of onsets fits the number of times subjects were presented with the task.

    Check for onset values that do not make sense, for example, onsets with no decimal places.

Check the maximum onset values. The maximum onset value can be checked against the expected number of scans or time of scanning for each subject. If the timing onsets are in scans, you would expect that the maximum value of the timing onsets not surpass the number of scans for the subject run. If the timing onsets are in scans and the maximum onset value is greater than the number of scans, it may indicate that either the scanner stopped early during the run, or the onsets are actually recorded in seconds.

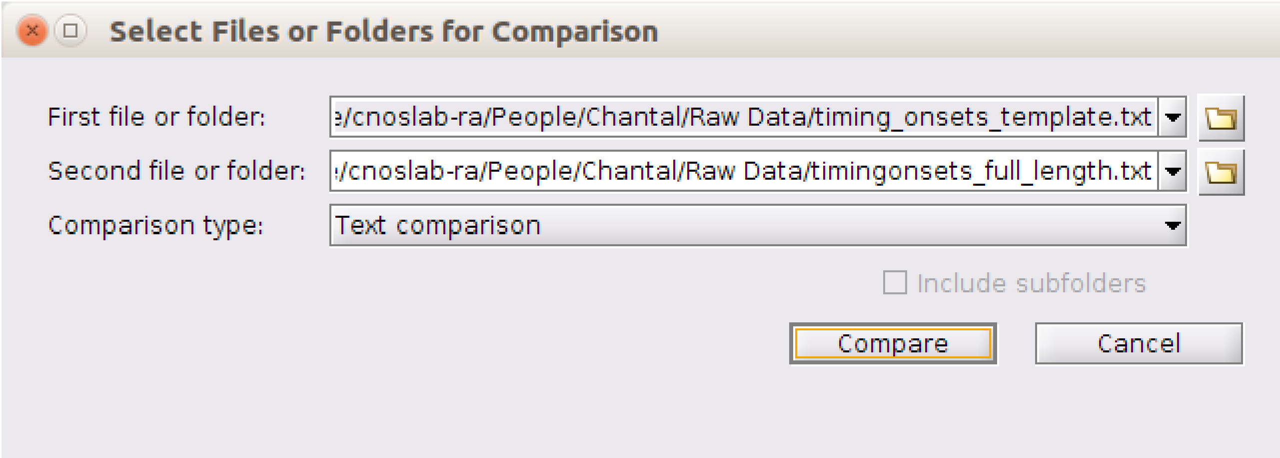
If you have timing onsets that surpass the number of scans, you will encounter a warning message when creating G. If you have already checked over your timing onsets, this message can be ignored.



Timing Onset Template Check

If you are using a previously created timing onset file, you should verify that it matches with the timing onsets template text file produced after editing Runs and Conditions. You can inspect the files however is easiest for you.

One way to view the two text files side by side is to use the Compare Against tool in MATLAB. Navigate to your directory where either of the onset text files are located. Right click on the onset text file and select “Compare Against”, then “Choose”. A window will appear and the path of the first file you selected should be located under “First file or folder”. Under “Second file or folder”, navigate to and open your other timing onset file. Leave “Comparison type” as Text comparison. Select “Compare” at the bottom.



This will open a window with both onset text files displayed side by side. Lines highlighted in red do not match. Lines beginning with a % are commented out, meaning that they do not have to match exactly for the onset file to be read correctly. Lines with the timing numbers will also be red as the template file will have empty parentheses while the completed onset file should have numbers. Verify that the number and order of subjects and runs are the same in both files.

```
                           [6 unmodified lines hidden]
% -------------------------------------- . % --------------------------------------
                                         .
% -------------------------------------- . % --------------------------------------
% --- timing onsets for subject n1360    x % --- timing onsets for subject 1 (n1360
% -------------------------------------- . % --------------------------------------
n1360_Phonologic = [ ];                  x n1360_functional_Condition_1 = [ 15.0117
n1360_Semantic = [ ];                    x n1360_functional_Condition_2 = [ 53.0057
                                         .
                                         .
% -------------------------------------- . % --------------------------------------
% --- timing onsets for subject n1903    x % --- timing onsets for subject 2 (n1903
% -------------------------------------- . % --------------------------------------
n1903_Phonologic = [ ];                  x n1903_functional_Condition_1 = [ 15.0998
n1903_Semantic = [ ];                    x n1903_functional_Condition_2 = [ 55.0540
                                         .
                                         .
% -------------------------------------- . % --------------------------------------
% --- timing onsets for subject n1926    x % --- timing onsets for subject 3 (n1926
% -------------------------------------- . % --------------------------------------
n1926_Phonologic = [ ];                  x n1926_functional_Condition_1 = [ 14.9023
n1926_Semantic = [ ];                    x n1926_functional_Condition_2 = [ 54.9427
                                         .
                                         .
% -------------------------------------- . % --------------------------------------
% --- timing onsets for subject n1958    x % --- timing onsets for subject 4 (n1958
% -------------------------------------- . % --------------------------------------
n1958_Phonologic = [ ];                  x n1958_functional_Condition_1 = [ 15.0202
n1958_Semantic = [ ];                    x n1958_functional_Condition_2 = [ 55.0138
                                         .
                                         .
% --------------------------------------   % --------------------------------------
```

Scree Plots

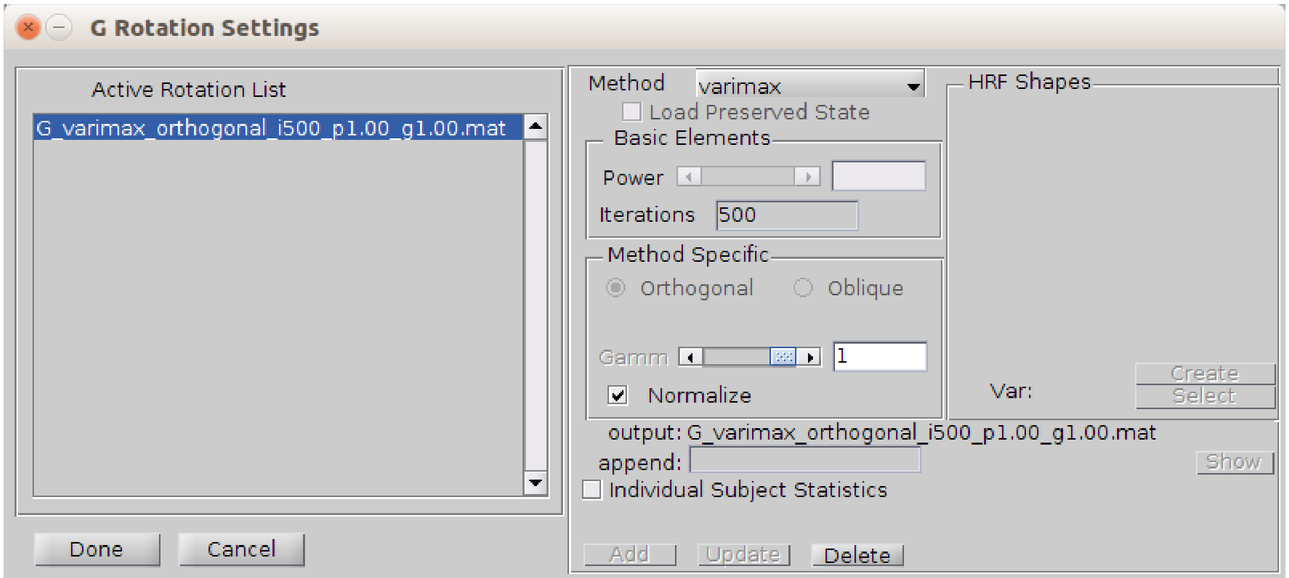
Verify with Todd how many components to extract based on the scree plot before proceeding.

HRFmax Rotation

Varimax rotation maximizes the variance between components. In hrfmax rotation, a shapes.mat file is created. This file contains the expected hdr shapes for the specific task timing. Hrfmax rotation finds the best *T* matrix for rotation of components that matches the shapes.mat file created. This method may produce the more biologically relevant rotation of components.

Instructions

1. Click on the GC+E button so that it is highlighted green.
2. If you have already extracted components, uncheck extract.
3. Check Rotate. Click the Settings button beside Rotate. This will open the G rotation Settings window.



4. Under Method, select hrfmax.

5. In the Basic Elements box, you can modify the Iterations field. This number is the number of T matrices that the program will search through to obtain the rotation of components that most matches your shapes.mat file. Enter 500,000 iterations for 2 or 3 component solutions, and 5,000,000 iterations for 4 or 5 component solutions. For 6 or more component solutions it is recommended to use Varimax rotation.

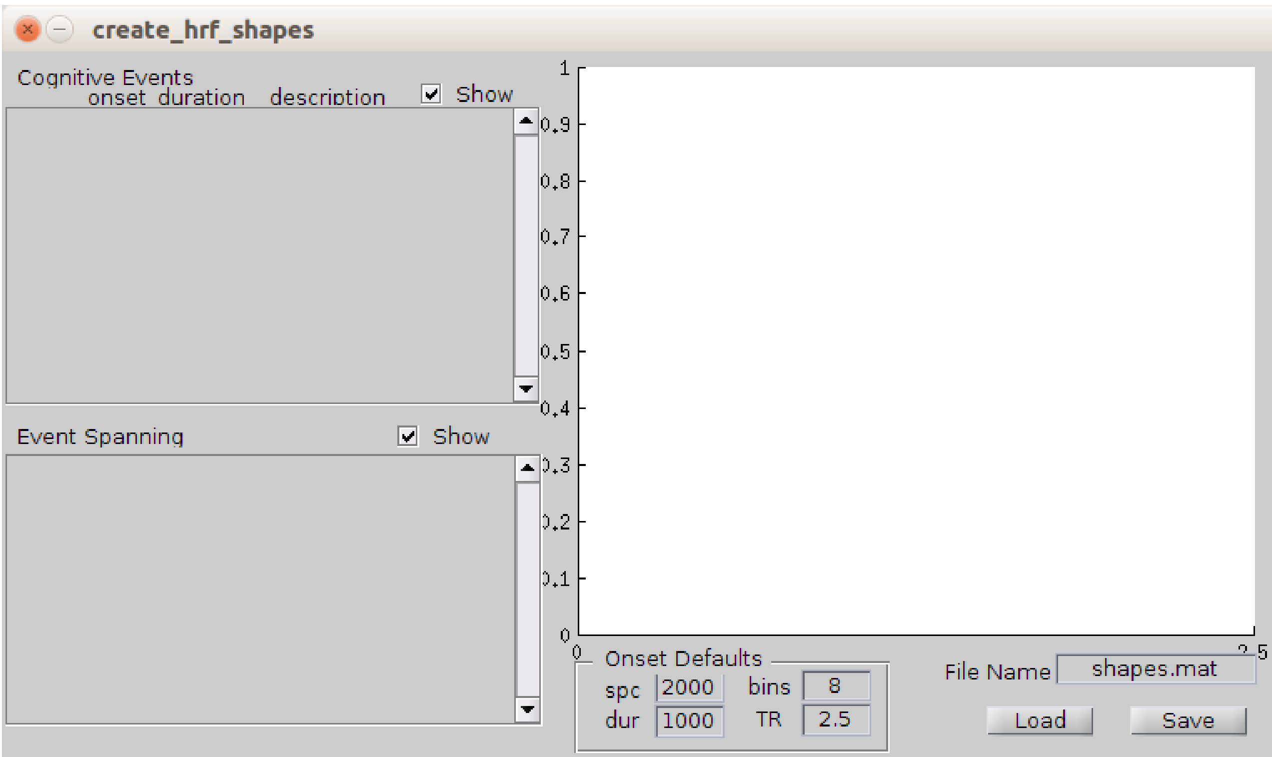
6. Select or create your shapes.mat file.

   If you have already created your shapes.mat file, click Select in the HRF Shapes box, and select your shapes.mat file.

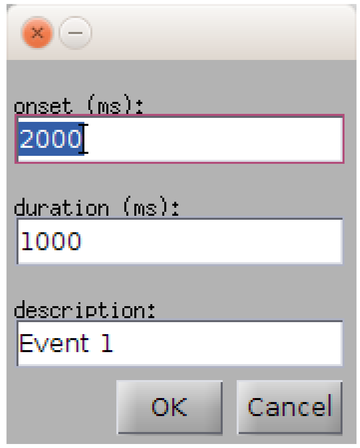
If you do not already have a shapes.mat file for your dataset, you will need to create one based on the hdr response corresponding to your task timing.

Create shapes.mat

- In the HRF Shapes box, click Create. This will open the create_hrf_shapes window.



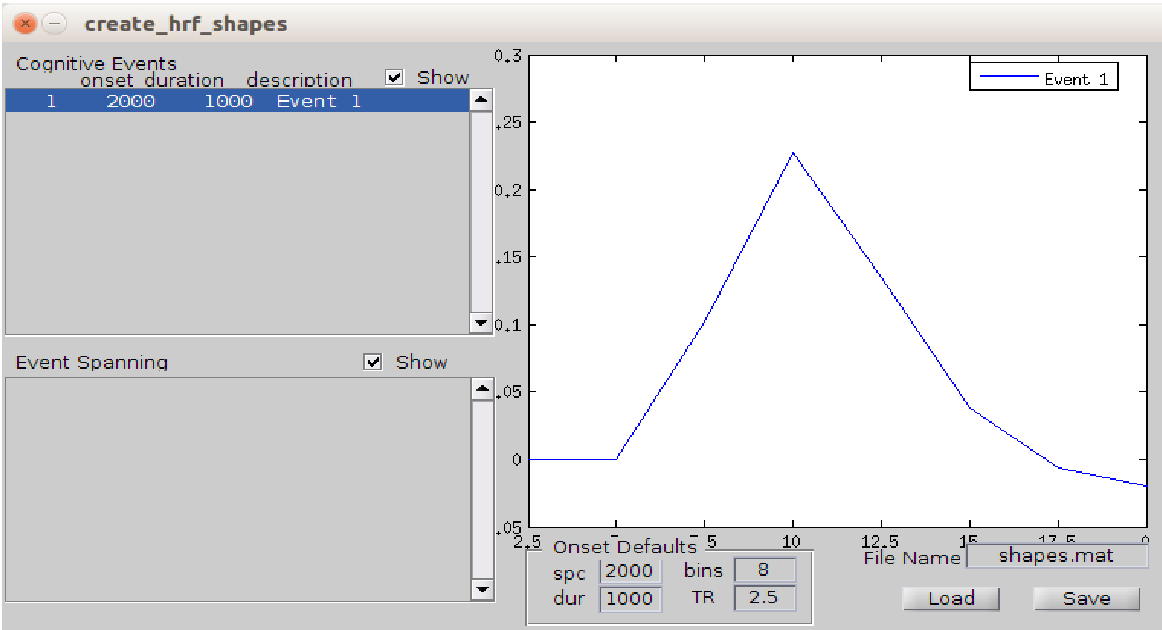
- In the Cognitive Events window, right click on the empty space. This will open a smaller window to create the hdr events.



- In the onset field, enter the beginning of your hdr response shape in milliseconds. Under the duration, enter the expected duration of the response in milliseconds. You can leave the description as Event 1. Click OK.
- You can create multiple hdr response events within the same create_hrf_shapes window by right clicking again in the space. The description field will update to Event 2, Event 3...
- For tasks that involve only one display and one response, the HRF shapes should typically be the following four shapes. Note that the values must be entered in milliseconds.

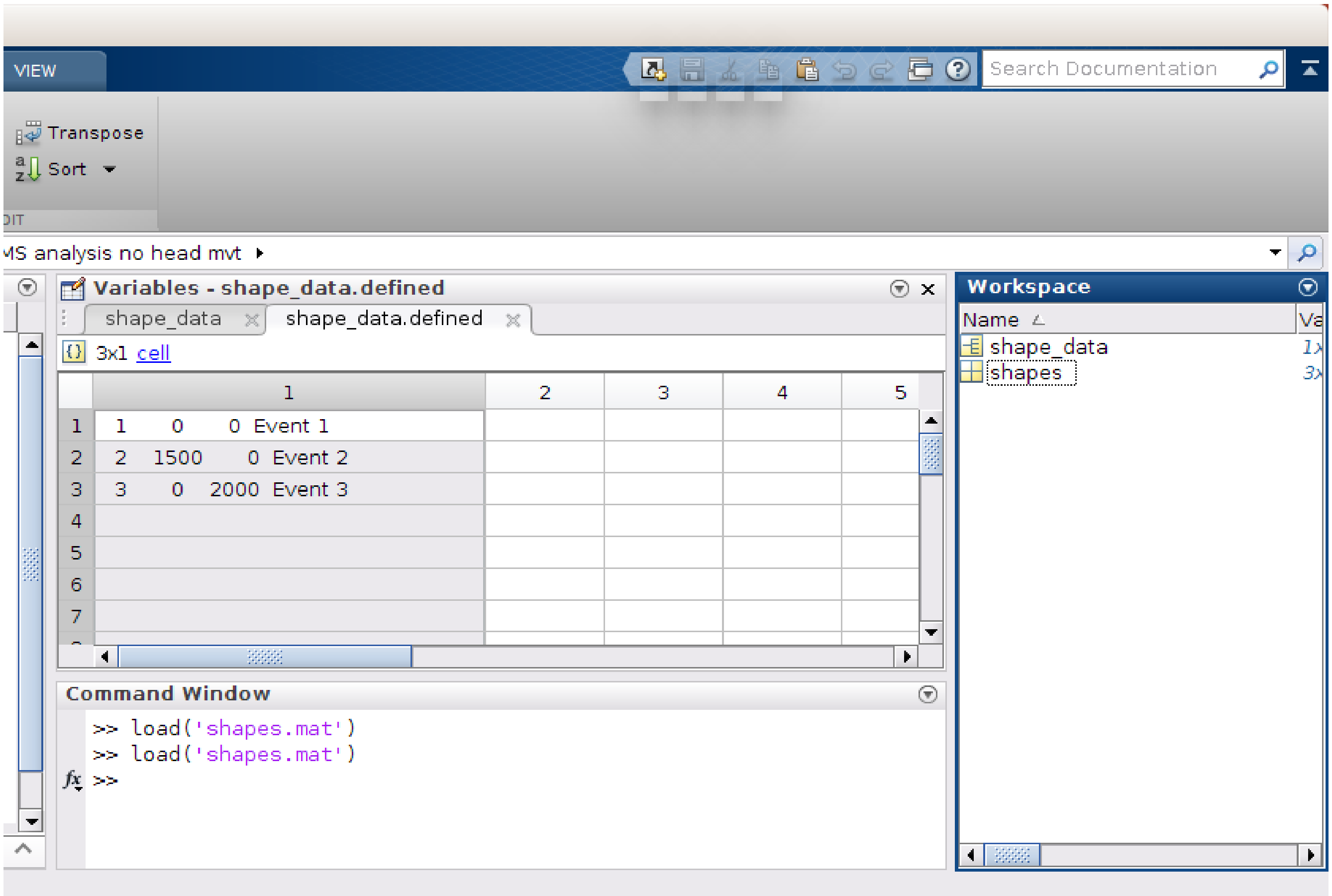
1) Event 1 (Visual Onset):
   - Onset= 0
   - Duration= 0
2) Event 2 (Visual Display):
   - Onset= 0
   - Duration= Screen Display (time stimuli remains on the screen)
3) Event 3 (Response Process):
   - Onset= Average reaction time for all groups
   - Duration= 0
4) Event 4 (Evaluation Process):
   - Onset= Average reaction time for all groups
   - Duration= 1500

- For tasks that do not involve only one display and one response, the HRF shapes used will have to be modified to correspond to task timing.
- The average reaction time may be calculated from your available data information. If you are unable to calculate the average reaction time or the information is not available, it is recommended to use 1000 ms for tasks with a display of 2 seconds or less.



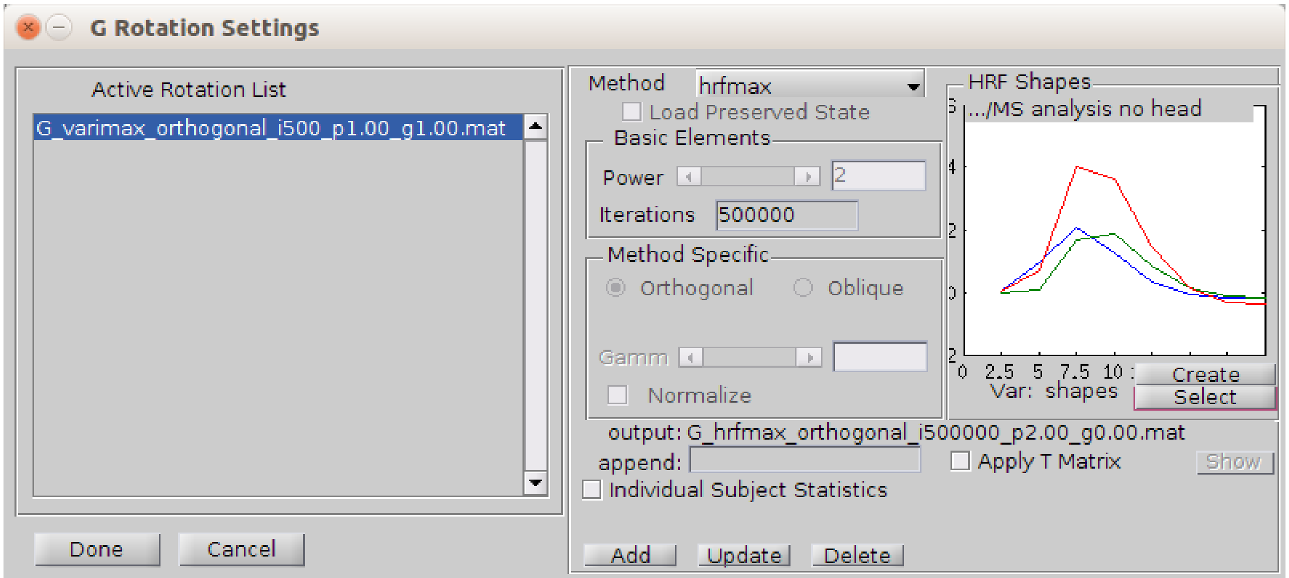
- As you create cognitive events, they will be plotted in the white box of the create_hrf_shapes window. When you have finishes creating events, select Save. This will save a shapes.mat file in your working directory.
- You can later verify you entered the correct events in the shapes.mat file in MATLAB. Open shapes.mat. Select Shape_data in your workspace. In the variables window select defined.

- This will open shape_data.defined which will display your events. The first number corresponds to inputted onset and the second number corresponds to inputted duration in milliseconds.
- **Verify the shapes with Todd before proceeding.**



7. After selecting or creating a shapes.mat file, ensure that the correct shapes are displayed in the

HRF Shapes box.



8. Verify that the Active Rotation List box has the rotations you would like to complete. You can add and remove rotations accordingly. To add hrfmax rotation, select the Add button at the bottom of the window. Verify that hrfmax is listed in the Active Rotation List. Since varimax was already completed in this example, it is deleted by highlighting it in the Active Rotation List and then selecting Delete and the bottom of the G Rotation Settings window.

9. Select Done in the G Rotation Settings window. This will bring you back to the main CPCA window. Ensure Rotation is checked and select RUN.

Merging Datasets for CPCA

Different datasets may be merged to further examine which functional brain networks are involved in specific tasks. Combining datasets with different experimental tasks allows us to look for networks that we would expect to appear, or not appear, in certain conditions in both experiments. This provides further insight into the function of the networks.

In order for two datasets to be merged, they should be preprocessed with the same version of SPM and parameters for normalization and smoothing. Ideally they will have the same TR, if not, the timing of the predictor weights in the HDR will need to be adjusted for each study for the plots.

Completing a merged analysis is similar to a regular analysis, with a few exceptions for data organization, Subject scans file list creation, and creation of the G matrix.

Data Organization

A merged analysis will require the subjects from both datasets to be found in the same data folder. You can use data linking to pull study data from various folders.

A consideration for merged dataset organization is that their data folders will need to match in structure. For example, if a dataset has a single run per subject, it may be organized as *Groups/Subjects/Scans* and not have a *Run* folder for the scans. This organization will work so long as both datasets being merged have this organization. However, should the other dataset have several runs and *Run* folders, and is organized as *Groups/Subjects/Runs/Scans,* the data organization will need to be modified for CPCA. This can be accomplishied with bash scripting to create a run folder and move all files into this folder for the dataset lacking run folders. Note that if you are using data links, changes made to the linked folders will also be made to the original folders and vice versa.

Merged datasets Timing Onsets

If you are merging studies, the timing onsets for all studies should all be in scans, or all be in seconds.

If the onsets are all in scans, input any study's TR into the G creation window. However, when plotting HDRs later, you will need to plot each study's predictor weights against their respective TR.

If the onsets are all in seconds, and all studies have the same TR, proceed as normal by selecting onsets in seconds and inputting the TR. However, if the studies have different TRs, you will need to convert all timing onsets from seconds to scans. To do so, divide all onsets in seconds by the study TR. Then run CPCA on the onsets for all studies in scans.

When creating the timing onsets template, do not click apply to all subjects and apply to all runs. Instead, select the conditions that pertain to each subject given the study they are from.

SAN Drive Backup

If you have completed your CPCA analysis off the SAN drive, be sure to copy over your results. As we have limited space on the SAN drive, only back up the files taht are required. If you need to back up files along the way, use an external hard drive as opposed to the SAN drive.

For analyses that may not be the final version: It is sufficient to simply copy over the *G* folder with the rotated component solution of choice from your analysis.

Once a final analysis has been selected: The full analysis should be backed up along with the smoothed fMRI scans. Ensure you leave a detailed README.txt in your folder on the lab computer and the SAN drive about where your files are located on the SAN drive, which location they originated from, and details about the file contents.